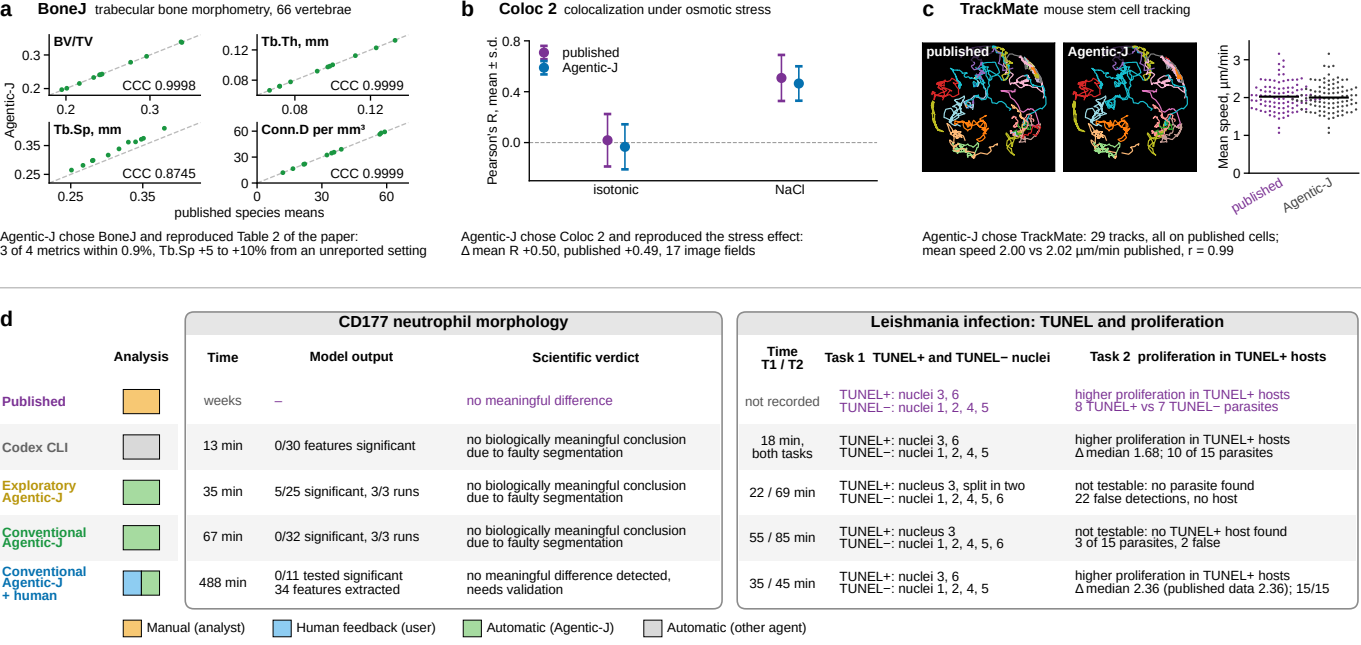



**d**

| Analysis | Time | Model output | Scientific verdict |
|---|---|---|---|
| **CD177 neutrophil morphology** | | | |
| Published | weeks | – | no meaningful difference |
| Codex CLI | 13 min | 0/30 features significant | no biologically meaningful conclusion due to faulty segmentation |
| Exploratory Agentic-J | 35 min | 5/25 significant, 3/3 runs | no biologically meaningful conclusion due to faulty segmentation |
| Conventional Agentic-J | 67 min | 0/32 significant, 3/3 runs | no biologically meaningful conclusion due to faulty segmentation |
| Conventional Agentic-J + human | 488 min | 0/11 tested significant 34 features extracted | no meaningful difference detected, needs validation |

| Time T1 / T2 | Task 1 TUNEL+ and TUNEL− nuclei | Task 2 proliferation in TUNEL+ hosts |
|---|---|---|
| **Leishmania infection: TUNEL and proliferation** | | |
| not recorded | TUNEL+: nuclei 3, 6<br>TUNEL−: nuclei 1, 2, 4, 5 | higher proliferation in TUNEL+ hosts<br>8 TUNEL+ vs 7 TUNEL− parasites |
| 18 min, both tasks | TUNEL+: nuclei 3, 6<br>TUNEL−: nuclei 1, 2, 4, 5 | higher proliferation in TUNEL+ hosts<br>Δ median 1.68; 10 of 15 parasites |
| 22 / 69 min | TUNEL+: nucleus 3, split in two<br>TUNEL−: nuclei 1, 2, 4, 5, 6 | not testable: no parasite found<br>22 false detections, no host |
| 55 / 85 min | TUNEL+: nucleus 3<br>TUNEL−: nuclei 1, 2, 4, 5, 6 | not testable: no TUNEL+ host found<br>3 of 15 parasites, 2 false |
| 35 / 45 min | TUNEL+: nuclei 3, 6<br>TUNEL−: nuclei 1, 2, 4, 5 | higher proliferation in TUNEL+ hosts<br>Δ median 2.36 (published data 2.36); 15/15 |

Manual (analyst) · Human feedback (user) · Automatic (Agentic-J) · Automatic (other agent)

